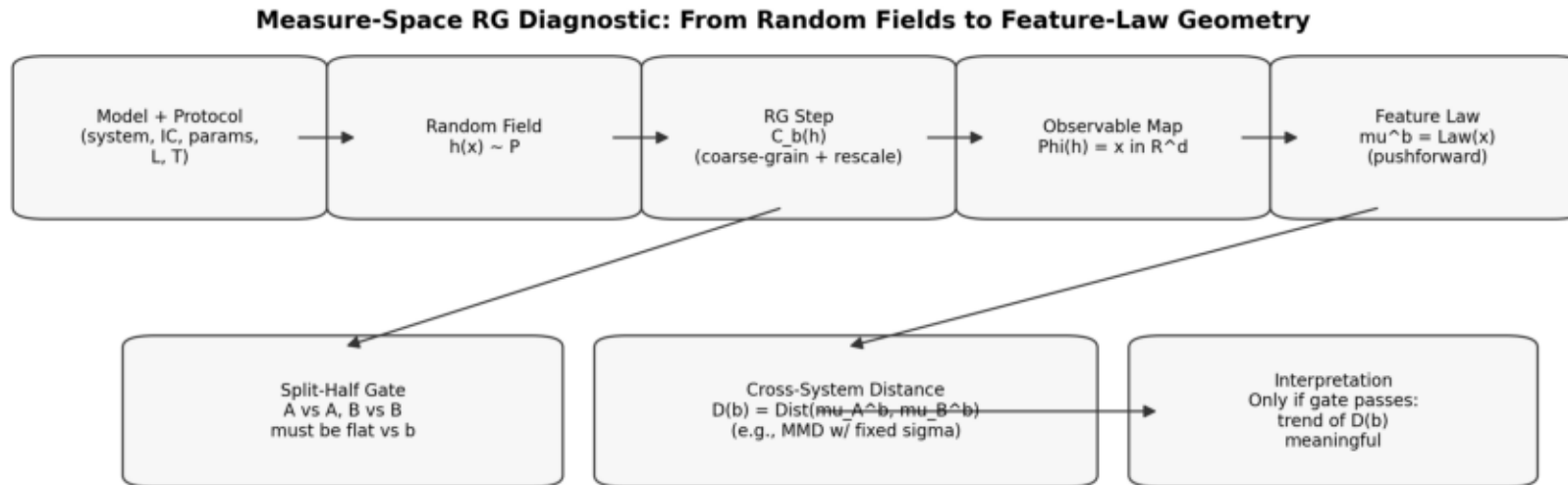
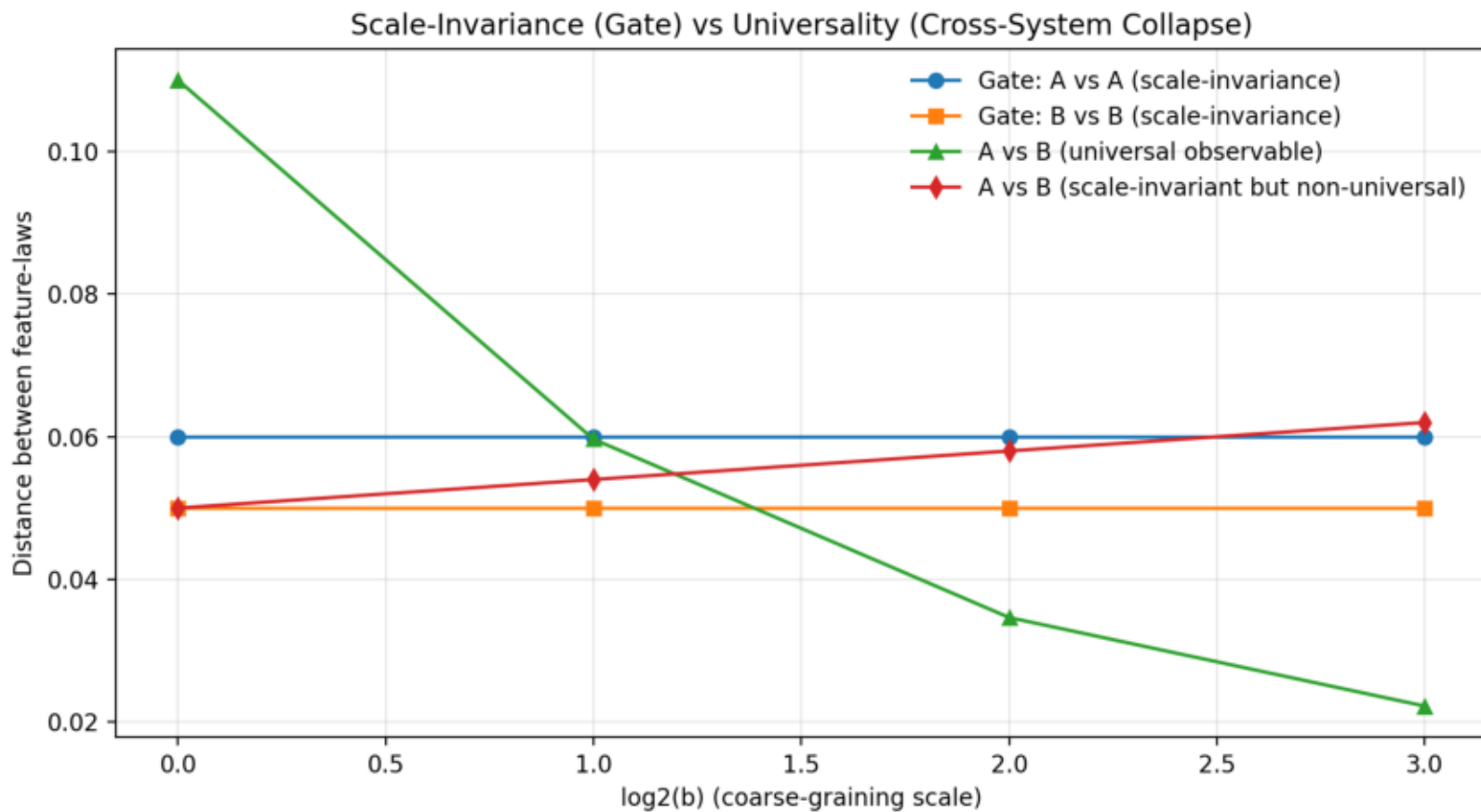


# Pipeline Schematic (Measure-Space RG + Diagnostic Gate)



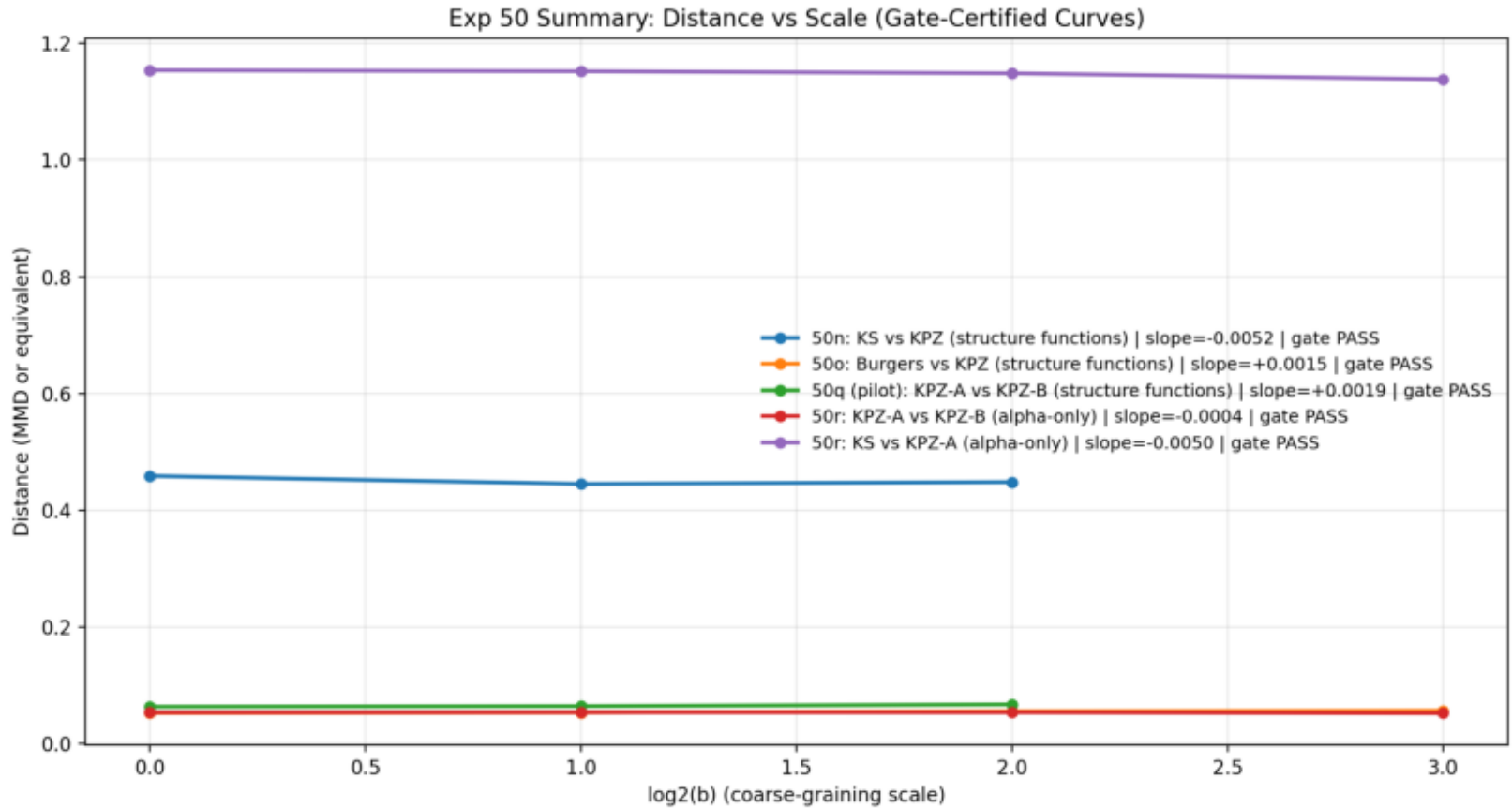
Conceptual: model -> random field law -> coarse-graining -> features -> feature-law; gate before interpretation.

# Scale-Invariance vs Universality (Conceptual)



Key lesson: passing the gate (self-consistency) does not guarantee cross-system collapse.

# Exp 50 Distance Summary (Gate-Certified Curves)

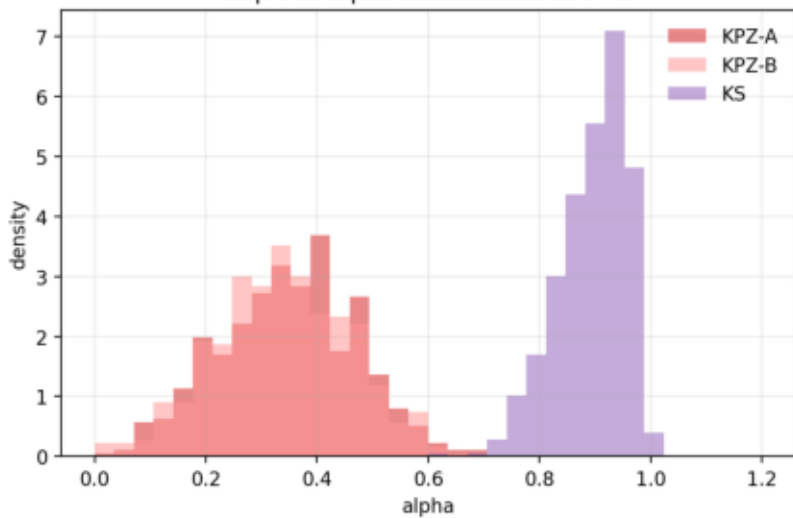


KS vs KPZ flat; Burgers vs KPZ non-collapse; KPZ vs KPZ depends on observable; alpha-only behaves as expected.

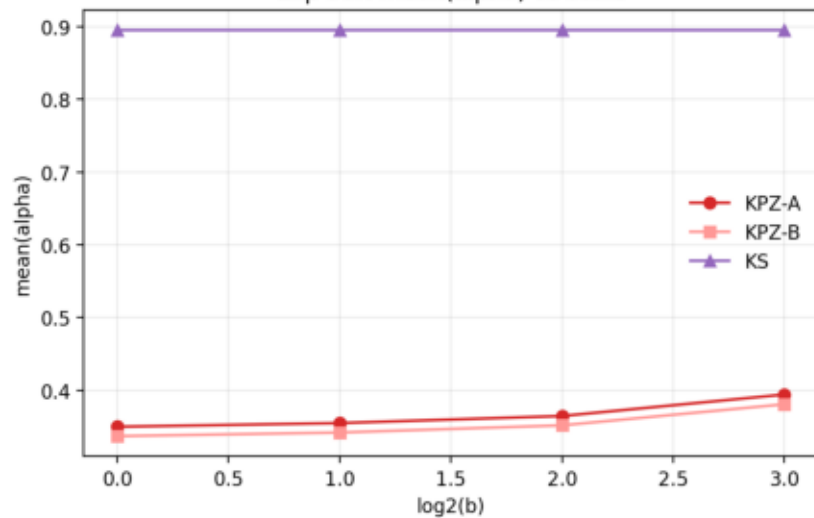
# Exp 50r: Alpha-Only Positive Control

Exp 50r Alpha-Only Positive Control (sigma=0.1160)

Exp 50r: alpha distributions at b=1



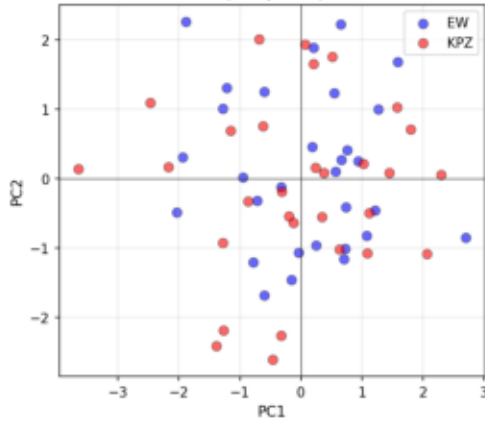
Exp 50r: mean(alpha) vs scale



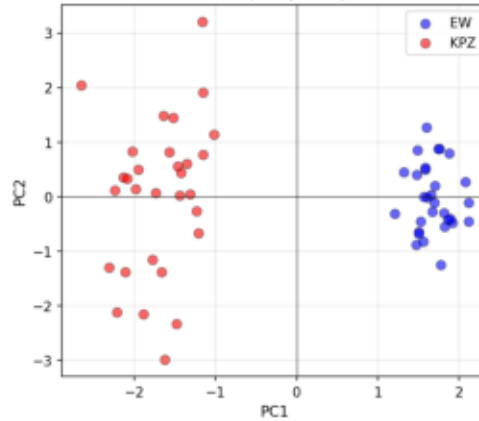
KPZ-A and KPZ-B alpha distributions overlap; KS is clearly separated.

# IC-Dependence (Exp 27)

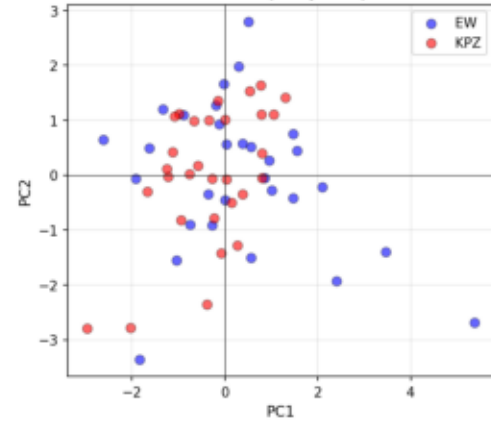
FLAT IC:  $r(\text{PC1}, \text{label}) = -0.060$



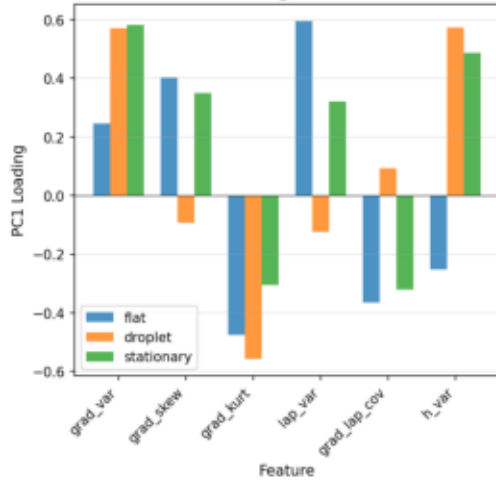
DROPLET IC:  $r(\text{PC1}, \text{label}) = -0.982$



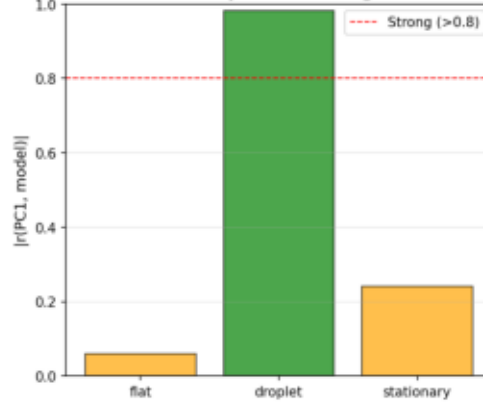
STATIONARY IC:  $r(\text{PC1}, \text{label}) = -0.241$



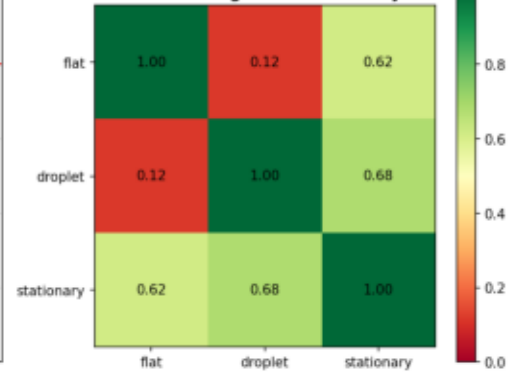
PC1 Loadings Across ICs



PC1 Separation Strength

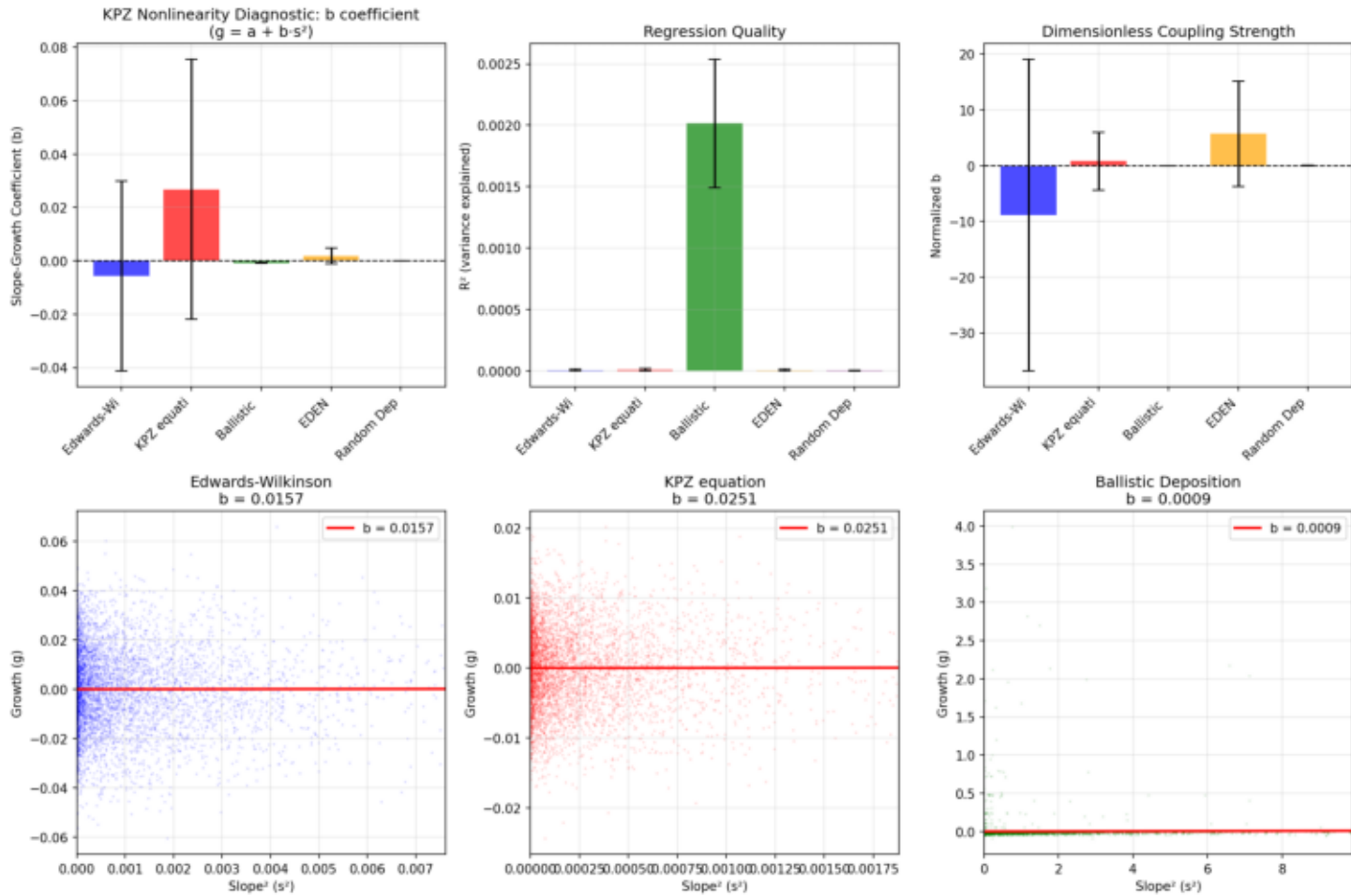


PC1 Loading Cosine Similarity



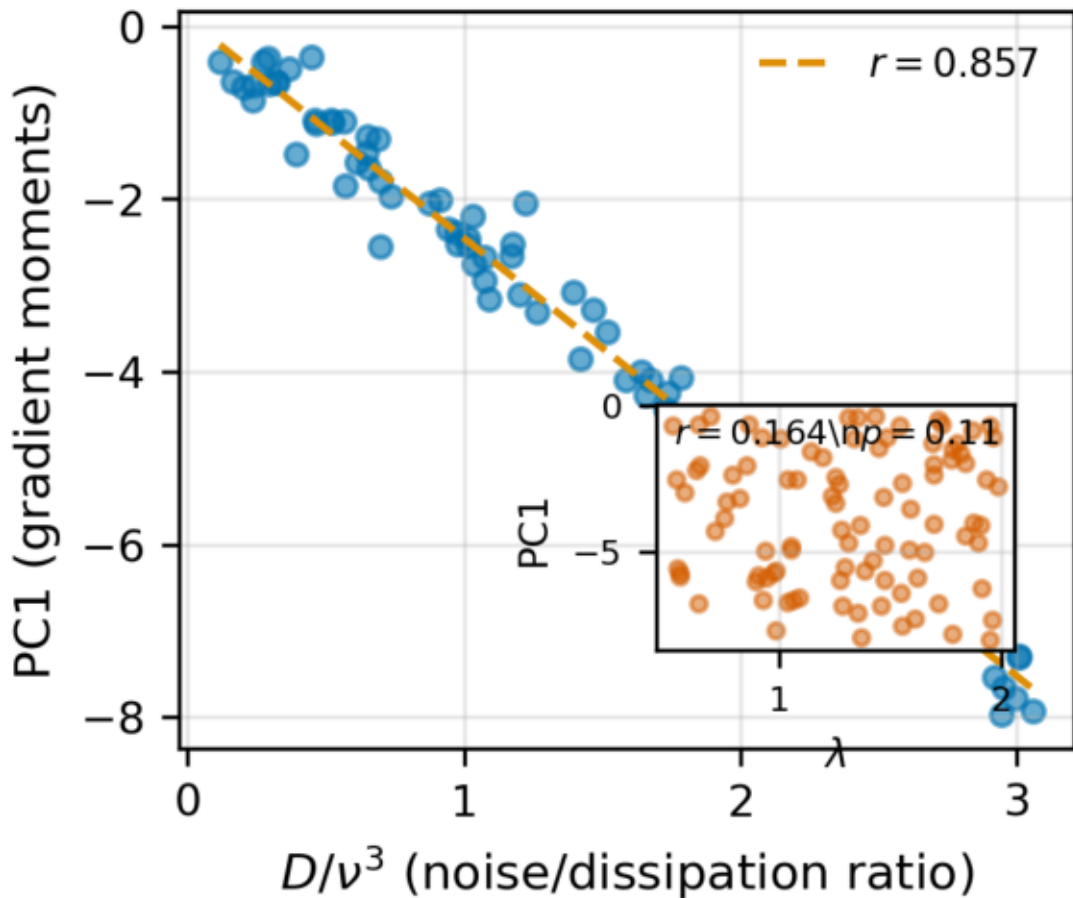
PC1 separation depends strongly on initial condition sector (flat vs droplet vs stationary).

# Slope-Growth Coupling (Exp 13)



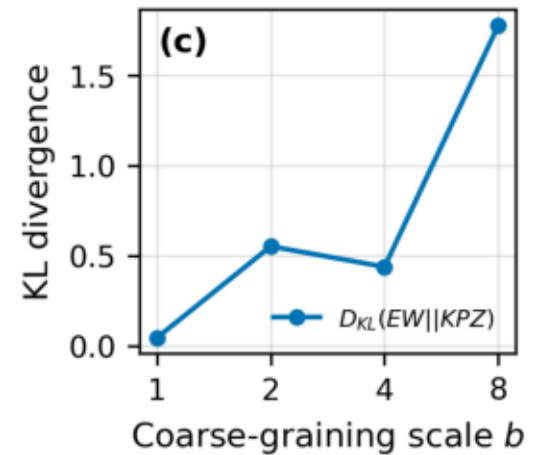
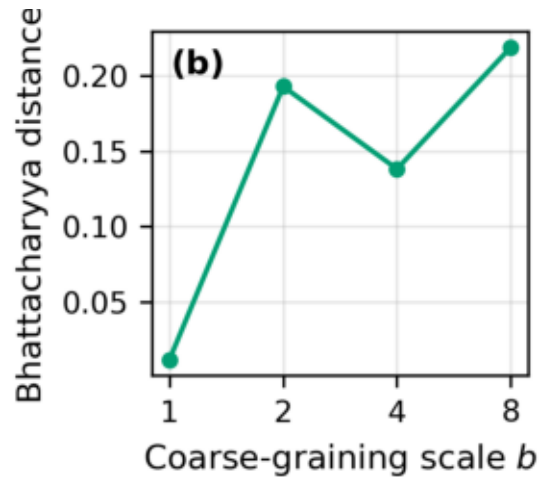
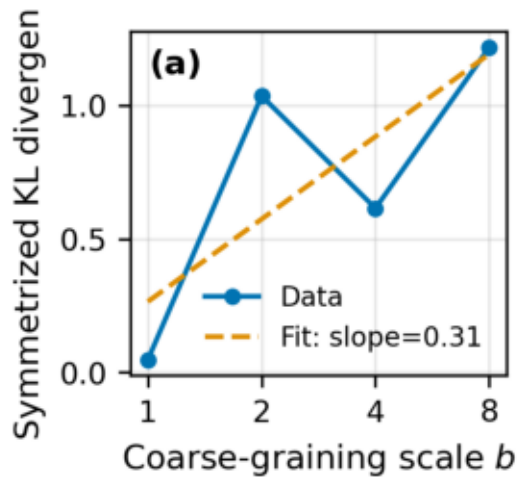
Direct physical diagnostic:  $\text{mean}(dh/dt)$  vs  $(\text{grad } h)^2$  differs for KPZ vs EW.

## Coupling Coordinate Discovery (Exp 46)



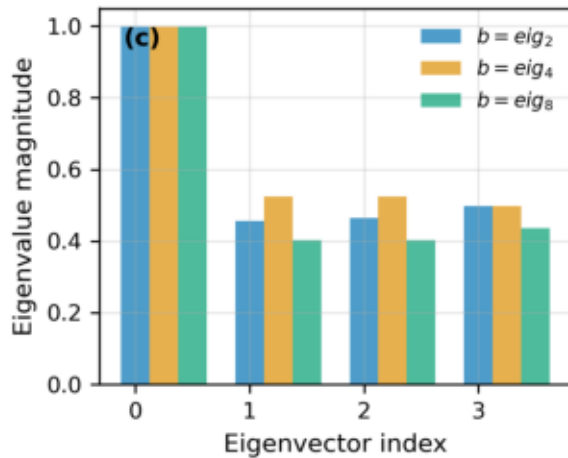
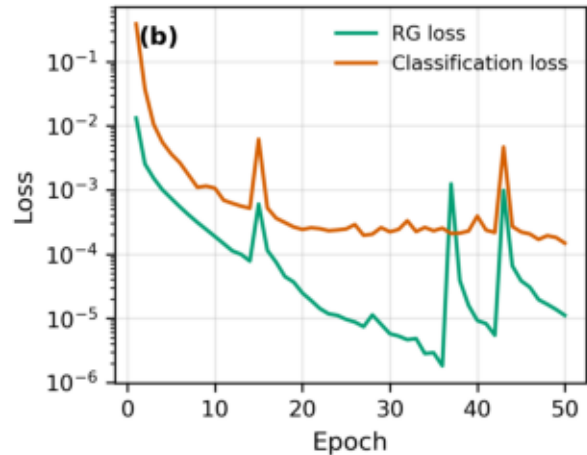
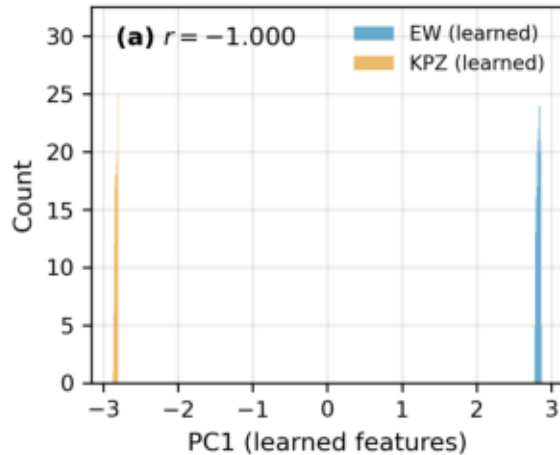
PC1 correlates with noise-to-dissipation ratio  $D/\nu^3$  across parameter sweeps.

# Information-Geometry Distances vs Scale (Exp 47)



KL / Bhattacharyya distances increase with coarse-graining: operational RG relevance.

# RG-Covariant Embedding (Exp 45b)



**(d)**

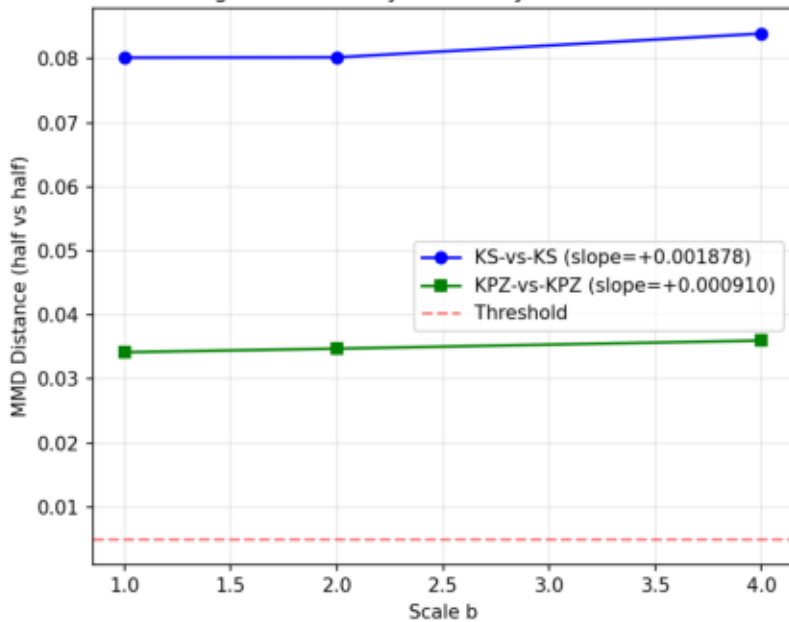
Validation Accuracy: 100%

Learned:  $r = -1.000$  Baseline:  $r = 0.990$

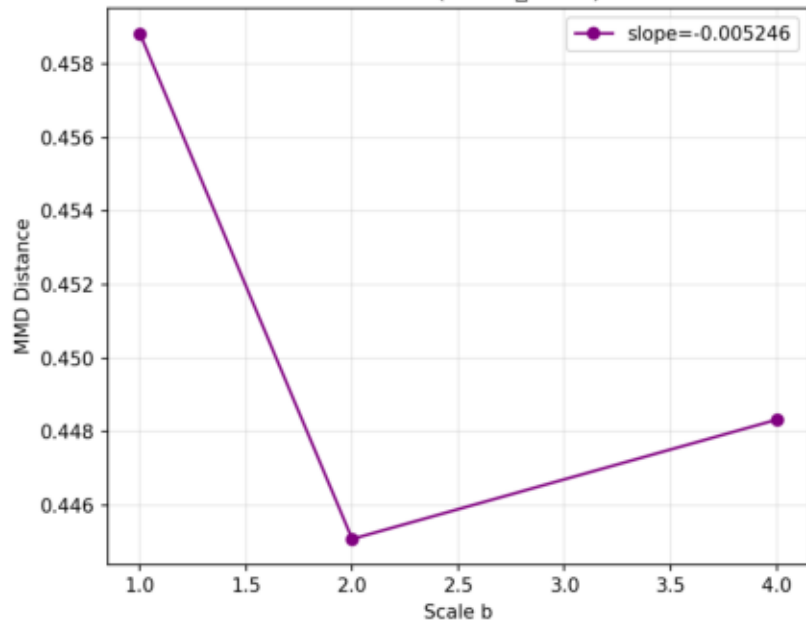
Learned features can be discriminative while obeying a scale-to-scale map (with anti-collapse constraints).

# KS vs KPZ: Scale-Free Structure Functions (Exp 50n)

Diagnostic Gate: System-vs-System Invariance



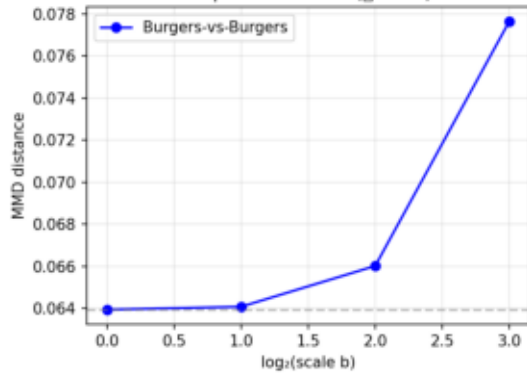
KS vs KPZ (Gate: ☒ PASS)



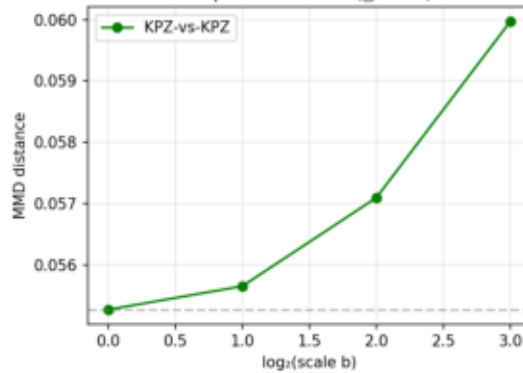
Gate passed; KS vs KPZ remains flat across scales (honest negative result).

# Burgers vs KPZ: Scale-Free Structure Functions (Exp 50o)

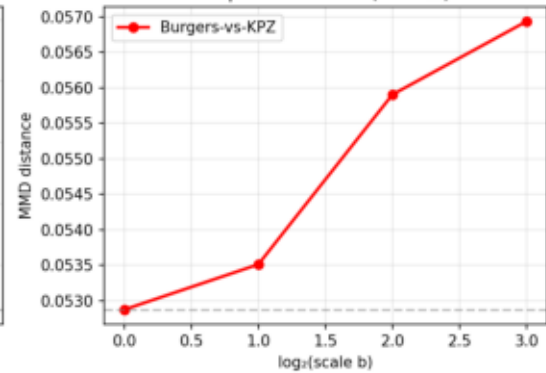
Diagnostic: Burgers-vs-Burgers  
Slope: +0.004300 (PASS)



Diagnostic: KPZ-vs-KPZ  
Slope: +0.001554 (PASS)

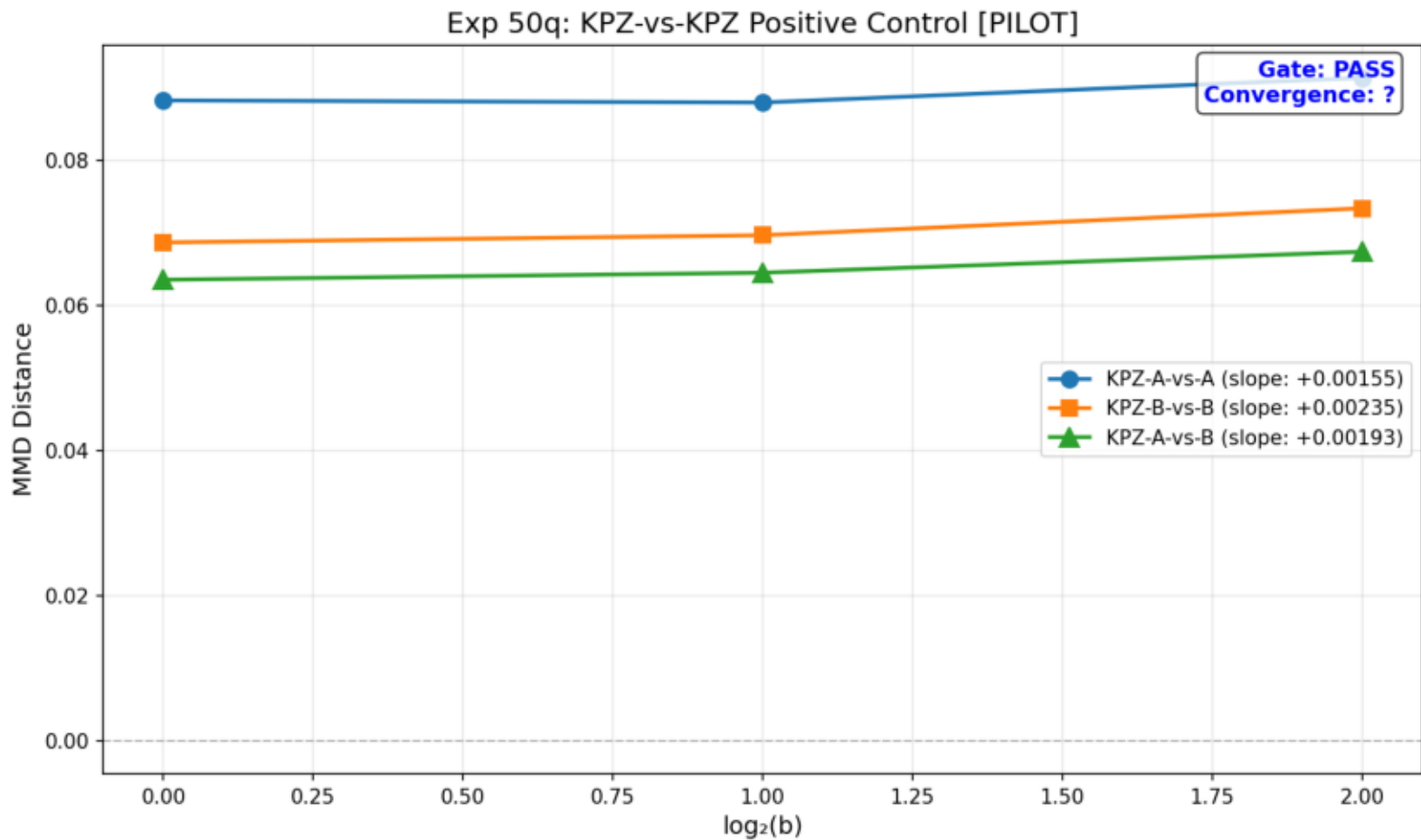


Burgers-vs-KPZ Distance  
Slope: +0.001456 (FLAT  $\Delta$ )



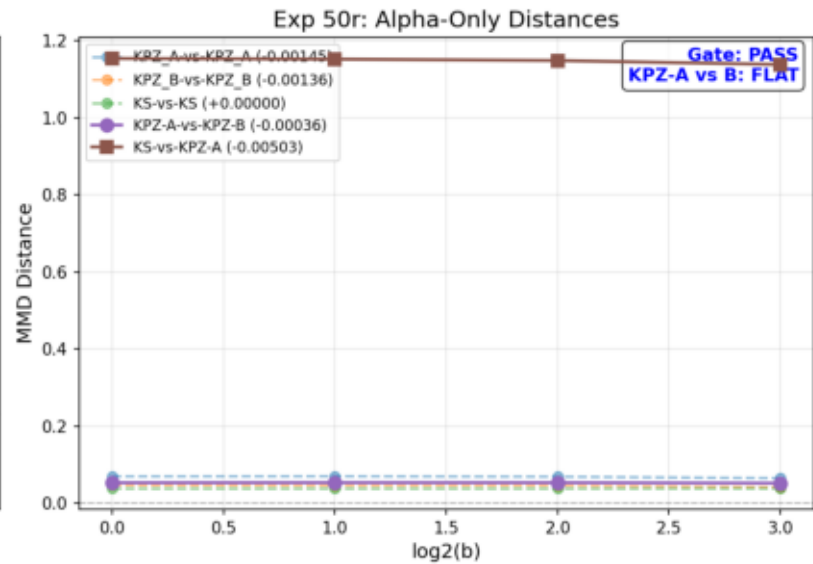
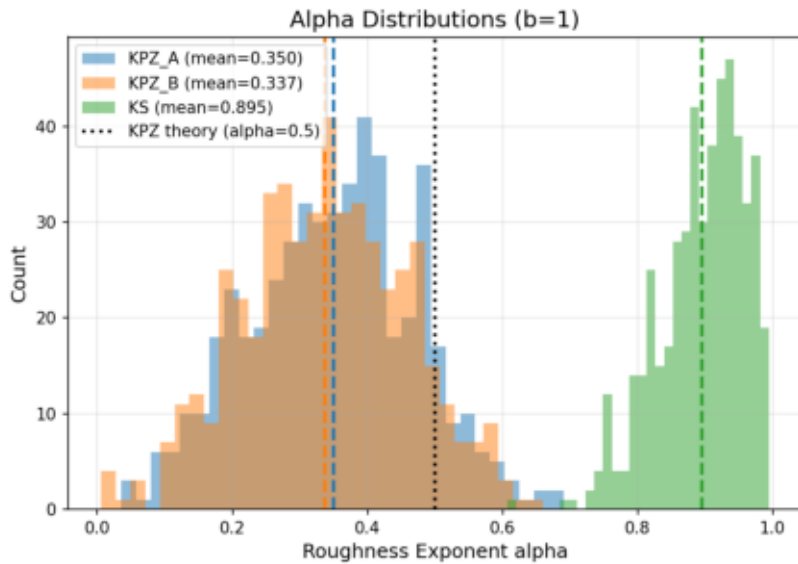
Gate passed; distance did not collapse (flags mismatch in positive control setup/observable).

# KPZ-A vs KPZ-B: Structure-Function Observable (Exp 50q pilot)



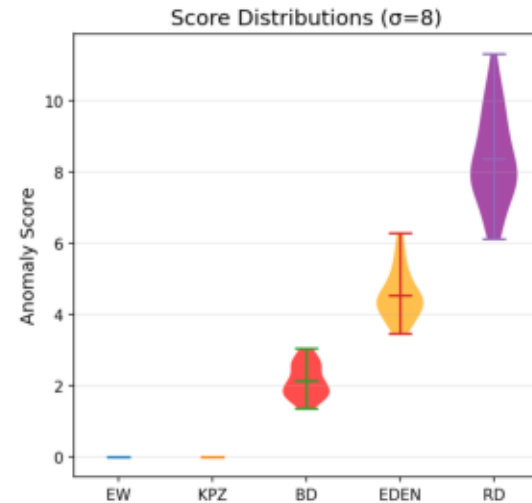
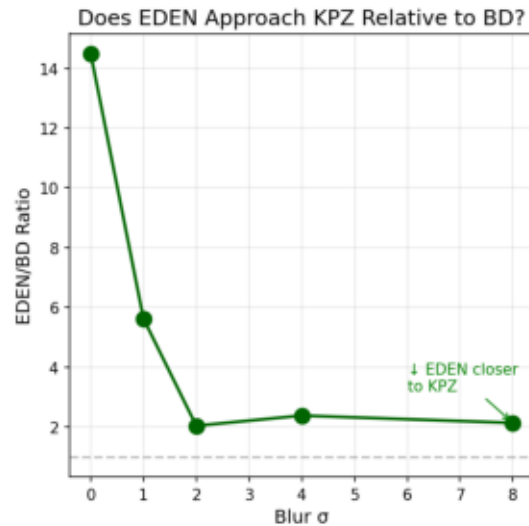
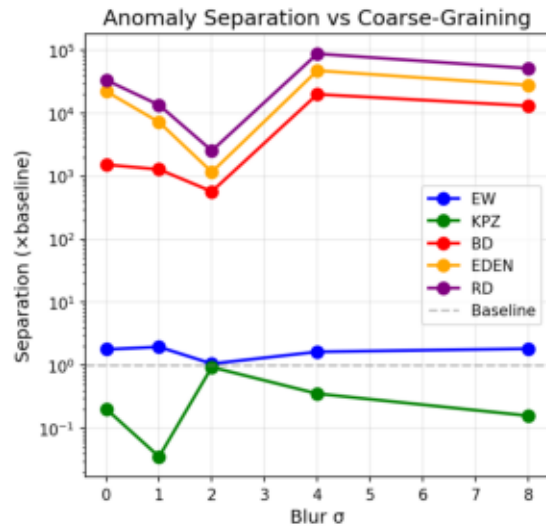
Gate passed; distance increased: scale-invariant but non-universal observable (finite-size/corrections).

# KPZ-A vs KPZ-B: Alpha-Only (Exp 50r)



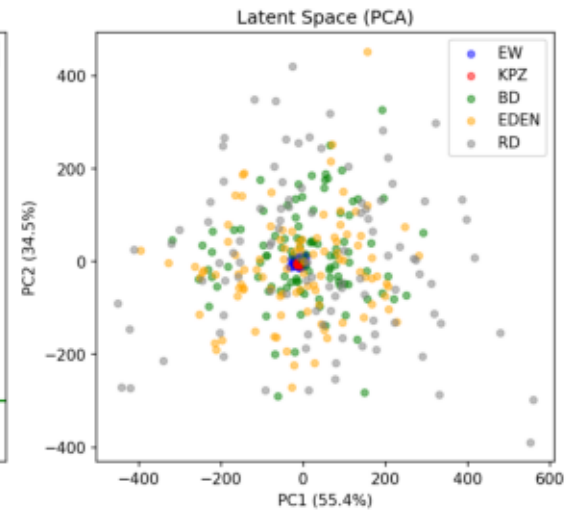
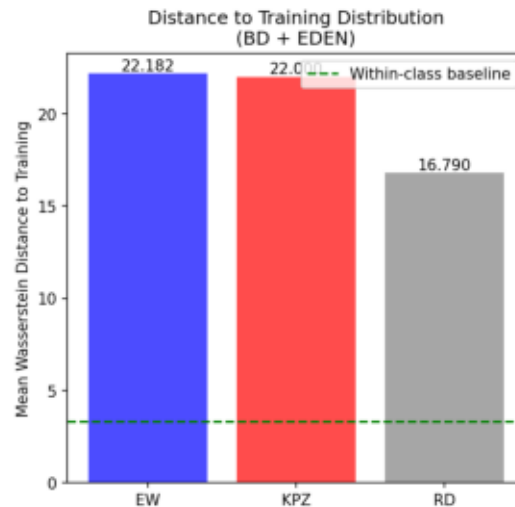
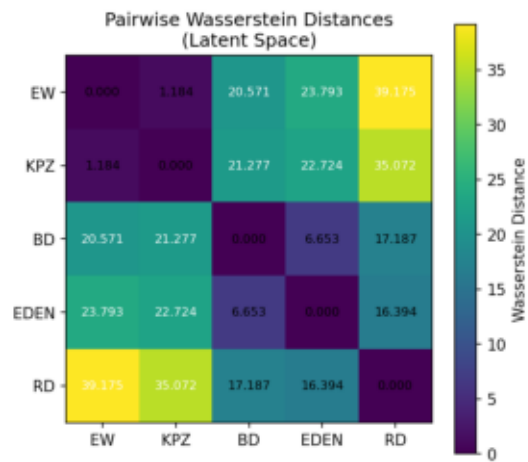
With a truly universal 1D feature ( $\alpha$ ), same-class systems overlap; KS separates.

# Early Geometry Snapshot (Exp 05)



How EW/KPZ organize in low-order gradient observable space (early empirical intuition).

# Wasserstein Distances (Exp 7b)



Distances in distribution space show class structure but can be dominated by implementation details.